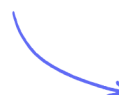


Structured Questions

Intermolecular Forces

Intermolecular Forces - Introduction / Hydrogen Bonding / Physical Properties / Solvent Choice

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Total Marks

/10

1 (a) This question is about the forces between molecules and ions.

Some data for three small molecules are shown.

Molecule	M_r	Boiling temperature / °C
Fluorine	38.0	-188
Hydrogen chloride	36.5	-85
Methanol	32.0	65

Explain the large variation in boiling temperatures, given the small range in M_r values.

Detailed descriptions of the forces involved are not required.

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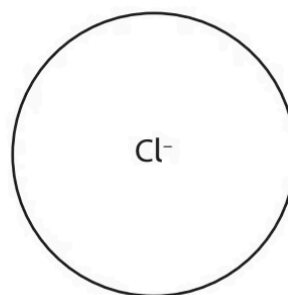
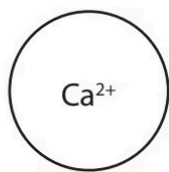
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(6 marks)

(b) Calcium chloride is soluble in water.

Complete the diagram to show how water molecules interact with each ion.
You may use  to represent a water molecule.



(2 marks)

(c) Explain why bromine is a liquid but iodine is a solid at room temperature.

Detailed explanations of the forces involved are not required.

(2 marks)